Class Interval	Tally Marks	Frequency
25-30	||	2
31-35	|||	3
36-40	||	2
41-45	𝍸 ||||	9
46-50	𝍸	5
51-55	𝍸 ||	7
56-60	𝍸 𝍸		11
61-65	𝍸		6
66-70	||	2
71-75			1
76-80	||	2
	TOTAL	**50**

Note

Consider any class, say 56- 60; then 56 is called the lower limit and 60 is called the upper limit of the class.

Progress Check

Form a frequency table for the following data:

23	44	12	11	45	55	79	20
52	37	77	97	82	56	28	71
62	58	69	24	12	99	55	78
21	39	80	65	54	44	59	65
17	28	65	35	55	68	84	97
80	46	30	49	50	61	59	33
11	57						

8.3 Measures of Central Tendency

It often becomes necessary in everyday life to express a quantity that is typical for a given data. Suppose a researcher says that on an average, people watch TV serials for 3 hours per day, it does not mean that everybody does so; some may watch more and some less. The average is an acceptable indicator of the data regarding programmes watched on TV.

Averages summarise a large amount of data into a single value and indicate that there is some variability around this single value within the original data.

A mathematician's view of an average is slightly different from that of the commoner. There are three different definitions of average known as the Mean, Median and Mode. Each of them is found using different methods and when they are applied to the same set of original data they often result in different average values. It is important to figure out what each of these measures of average tells you about the original data and consider which one is the most appropriate to calculate.

8.4 Arithmetic Mean

8.4.1 Arithmetic Mean-Raw Data

The Arithmetic Mean of a data is the most commonly used of all averages and is found by adding together all the values and dividing by the number of items.

For example, a cricketer, played eight (T20) matches and scored the following scores 25, 32, 36, 38, 45, 41, 35, 36.

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